

Barefoot to the Scene: Forensic Friction-Ridge Analysis of a Vandalism Case

Comparative ACE-V Examination of Crime-Scene Impression CFA1296 Against Three Reference Impressions

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SCENARIO — Partial barefoot impression (CFA1296) recovered from wet concrete at a vandalism site | Compared against three suspect reference impressions: Jo Ronaldo (CFA1297), Taylor Messi (CFA1298), Casey Mbappe (CFA1299)

1 Introduction & Background

Scenario. A partial barefoot friction-ridge impression (CFA1296) was found in wet concrete near damaged equipment at a vandalism site. Barefoot impressions were provided by three people: Jo Ronaldo (CFA1297), Taylor Messi (CFA1298) and Casey Mbappe (CFA1299).

In Australia, plantar ridges are analysed through the ACE-V principle to assist in identification/exclusion.⁷ However, reliability depends on print quality, substrate distortion, and examiner interpretation. Evidential weight is increasingly a requirement of Australian courts,⁸ and is expressed as a likelihood ratio (LR). This analysis compares CFA1296 to the three impressions, calculates an LR, and critically evaluates the results with consideration to the vandalism scenario.

Objective

To compare CFA1296 to three reference prints, calculate a likelihood ratio, and critically examine the limitations of plantar friction-ridge analysis in this context.

Method — ACE-V

- Analysis (CFA1296):** partial forefoot impression preserving the distal metatarsal/phalangeal area. Clear core and delta visible; ridges curve inward, indicating a right forefoot. 12 intervening ridges between core and delta; some degradation but sufficient detail for comparison.
- Comparison:** each reference print compared at Level 1 (pattern, core/delta geometry, ridge flow). Level 2 conducted only where Level 1 was compatible, following the first 10 ridges from the core.
- Exclusion:** inconsistent Level 1 prints excluded; consistent prints advanced to Level 2 evaluation of minutiae type and spatial arrangement.
- Verification:** a second examiner is required for operational reporting.

4 Conclusion

The first two suspects (Messi and Mbappe) were excluded at Level 1. Jo Ronaldo was **not excluded** as the source of CFA1296 (moderate LR $\approx$ 69). However, the concrete substrate caused degradation and distortion of the crime-scene impression. The calculated LR is a measure of evidential weight, not proof. Given the partial impression and the subjective components of ACE-V, footprint evidence must be treated as associative intelligence — not proof of identity. Independent verification and integration with other investigative evidence remain vital to a sound forensic determination.

2 Analysis & Results

Laterality: CFA1296 is a right forefoot. Core and delta both visible; ridge count between them = 12. Tracing ridges 1–10 from the core identified minutiae on ridges 1 (End), 3 (Bifurcation), 5 (End), 7 (Dot) and 9 (End); no notable minutiae on ridges 2, 4, 6, 8 or 10.

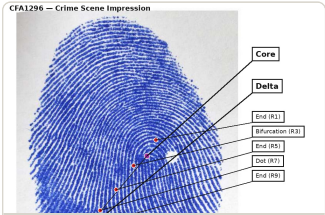


Fig. 1. CFA1296 (crime-scene impression) annotated with core, delta, and Level 2 minutiae within the first 10 ridges. Chain of custody maintained.

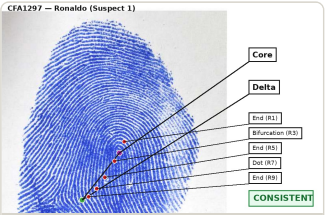


Fig. 2. CFA1296 vs CFA1297 (Ronaldo). Level 1 & 2 consistent; not excluded as source.

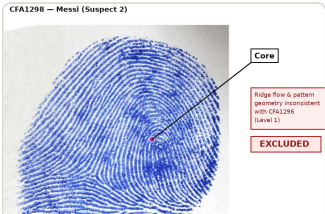


Fig. 3. CFA1296 vs CFA1298 (Messi). Ridge count 8 vs 12; Level 1 inconsistent — excluded.

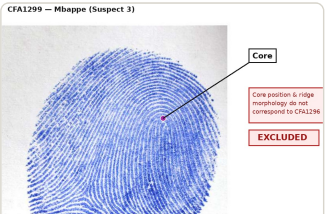


Fig. 4. CFA1296 vs CFA1299 (Mbappe). Non-corresponding core/ridge morphology — excluded.

Ronaldo
NOT EXCLUDED

Messi
EXCLUDED

Mbappe
EXCLUDED

Likelihood Ratio Calculation (frequencies from minutiae reference table)
End 63.2% · Bifurcation 38.2% · Dot 15.1%

0.632 × 0.632 × 0.632 × 0.382 × 0.151 = 0.0145

LR = 1 / 0.0145 $\approx$ 69 — supports Ronaldo (CFA1297) over an unrelated person.

3 Interpretation & Evaluation

Interpretation

Despite Ronaldo's denial, only his impression was not excluded through ACE-V. This is *not* proof of vandalism — only a print discovered at the scene; it does not establish timing or directly link him to the damage. This is strong investigative direction, but must be supplemented with other evidence and confirmed independently before any finding of guilt.

Evaluation & Limitations

- Combined frequency (0.0145; LR $\approx$ 69) uses the product rule, assuming minutiae independence and ignoring spatial configuration.
- Wet concrete caused unequal compression, distortion, and detail loss from movement during setting.⁶ Plantar measurements are affected by substrate distortion.³
- ACE-V remains somewhat subjective;⁵ Australian examiner agreement is 94.8%, though disagreement rises in suitability cases.¹
- Population frequencies vary by demographic/biomechanical factors — an Australian-specific database would strengthen the statistics.⁴
- LR reporting was transparent, consistent with recommendations for defensible friction-ridge evidence in Australian courts.⁴

References

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